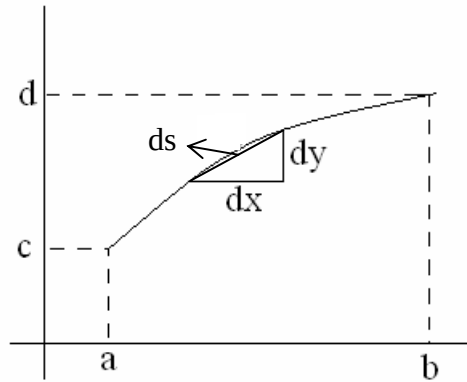


§8-1 Arc Length

Homework : 5,9,11,17,29,37

公式：弧長



$$\begin{aligned}
 l &= \int_a^b ds \\
 &= \int_a^b \sqrt{(dx)^2 + (dy)^2} \\
 &= \begin{cases} \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx \\ \int_c^d \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} dy = \int_c^d \sqrt{1 + (g'(y))^2} dy \end{cases}
 \end{aligned}$$

Example1 : $y^2 = x^3$ (1,1) to (4,8).

Solution :

$$\begin{aligned}
 y &= x^{\frac{3}{2}} \\
 l &= \int_1^4 \sqrt{1 + \frac{9}{4}x} dx \\
 &= \frac{3}{2} \left(1 + \frac{9}{4}x \right)^{\frac{3}{2}} \Big|_1^4.
 \end{aligned}$$

Example2 : $y = x^2 - \frac{1}{8} \ln x$ (1,1) to (x,y).

Solution :

$$\begin{aligned}y' &= 2x - \frac{1}{8x} \\ \Rightarrow l &= \int_1^x \sqrt{1 + \left(2t - \frac{1}{8t}\right)^2} dt \\ &= t^2 + \frac{1}{8} \ln |t| \Big|_1^x \\ &= x^2 + \frac{1}{8} \ln |x| - 1\end{aligned}$$

Example3 : Find the length of the curve $y = \int_1^x \sqrt{t^3 - 1} dt$, $1 \leq x \leq t$.

Solution :

$$\begin{aligned}y' &= \sqrt{x^3 - 1} = f'(x) \\ \Rightarrow l &= \int_1^4 \sqrt{1 + x^3 - 1} dx \\ &= \frac{2}{5} x^{\frac{5}{2}} \Big|_1^4 \\ &= \frac{62}{5}\end{aligned}$$

Example4 : Prove that the circumference of the circle with radius r is $2\pi r$.

Prove :

$$\begin{aligned}y = f(x) &= \sqrt{r^2 - x^2} \Rightarrow f'(x) = -x(r^2 - x^2)^{-\frac{1}{2}} \\ \Rightarrow 1 + (y')^2 &= \frac{r^2}{r^2 - x^2} \\ \Rightarrow \text{圓周長} &= 4r \int_0^r \frac{1}{\sqrt{r^2 - x^2}} dx \\ &= 4r \int_0^{\frac{\pi}{2}} \frac{1}{r \cos \theta} r \cos \theta d\theta \\ &= 2\pi r.\end{aligned}$$